

3D Fluid Sim

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Abstract

We are building a real-time 3D fluid simulation that produces visually realistic water using CUDA for physics and OpenGL for rendering. Our goal is to have a live demo of the sim at the end of the semester and gain hands-on experience multithreaded programming during the development process

1 Introduction

What project problem you or your team is trying to answer

2 Background

Why this problem is important

3 Approach

We will implement a real-time 3D fluid simulation using CUDA for physics and OpenGL for rendering. CUDA kernels will compute advection, pressure projection, and free-surface tracking, while OpenGL will handle rendering and shading. The CPU will coordinate simulation steps and input.

3.1 Resources Needed

- **Hardware:** NVIDIA CUDA-capable GPU, multi-core CPU.
- **Software:** CUDA toolkit, OpenGL, Nsight.

3.2 Uncertainties

- Pressure solver performance (Jacobi vs. Conjugate Gradient/Multigrid)
- Surface method (marching cubes vs. raymarching)
- CUDA optimization (memory layout, block/grid configuration)
- Balancing realism with visual quality

3.3 Task Division (4 People)

- **Member A:** Velocity field and scalar advection kernels
- **Member B:** Pressure solver, projection, divergence computation
- **Member C:** Free-surface (level set) tracking, surface extraction
- **Member D:** Rendering pipeline, water shaders, UI, and demo integration

All members will collaborate on debugging, performance optimization, and final demo preparation.

4 Evaluation

Explain your evaluation setup and results.

5 Related Work

None right now

6 Conclusion

Timeline of the project, using deadlines from the project description.

1. September 5th – Project groups must be set.
2. September 19th – Proposal due.
3. October 15th – Design Document due: we will clearly outline the technical implementation plan from start to finish, and detail potential hurdles with strategies for addressing them.
4. October 31st – Implementation and Demo Checkpoint I: display API completed and a working 2D fluid simulation.
5. November 30th – Implementation and Demo Checkpoint II: full 3D simulation implemented.
6. December 2nd and December 4th – Project presentations in class: present the full 3D simulation, demonstrate user interaction, explain key implementation details, and answer questions.
7. December 5th – Final project report due.

References